

Exp :5 Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

Aim

To implement the **K-Nearest Neighbours (K-NN)** algorithm in Python and classify test data using an appropriate dataset.

Algorithm

1. Load the dataset.
2. Split the dataset into training and testing sets.
3. Choose the value of **K**.
4. Train the K-NN classifier using the training data.
5. Predict the class labels for the test data.
6. Evaluate the model using accuracy and confusion matrix.
7. Display the results.

Program :

```
# Implementation of K-Nearest Neighbours (K-NN)

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load Iris Dataset

iris = load_iris()

X = iris.data

y = iris.target

# Split dataset into training and testing data

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

# Create KNN classifier with K = 3

knn = KNeighborsClassifier(n_neighbors=3)

# Train the model
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knn.fit(X_train, y_train)

# Predict test data
y_pred = knn.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

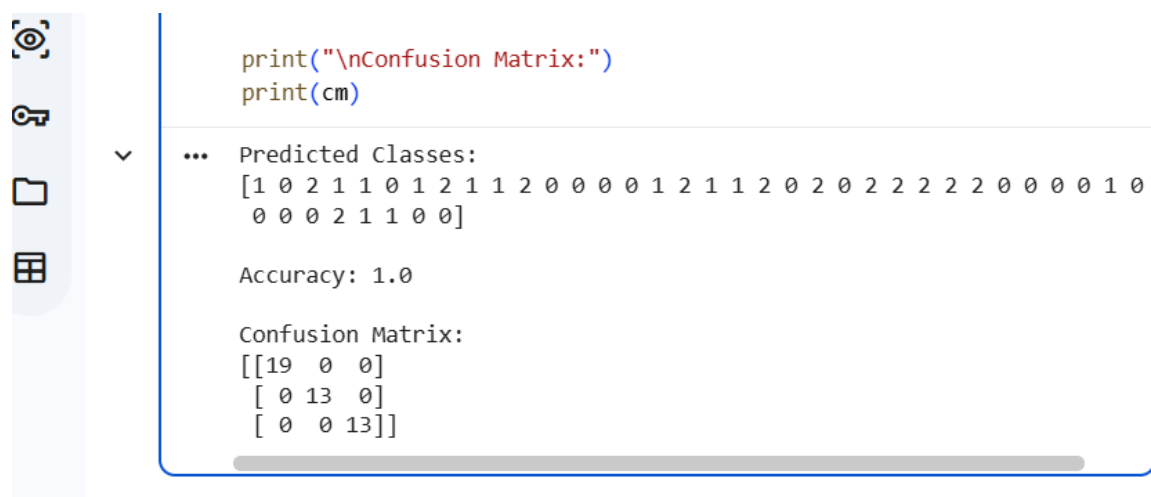
# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("Predicted Classes:")
print(y_pred)

print("\nAccuracy:", accuracy)

print("\nConfusion Matrix:")
print(cm)

```



```

print("\nConfusion Matrix:")
print(cm)

```

... Predicted Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0
0 0 0 2 1 1 0 0]

Accuracy: 1.0

Confusion Matrix:
[[19 0 0]
[0 13 0]
[0 0 13]]

Result

The **K-Nearest Neighbours (K-NN)** algorithm was successfully implemented in Python using the Iris dataset. The classifier was trained and tested, and the classification accuracy was obtained successfully